

# Math 571 - Homework 2

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Problems numbered [B:2.1:3] are from [Bergman's exercise notes](#). Problems labeled [R:2.1] are directly from Rudin. If I add an asterisk (\*) to a problem, we have a slight variant of the problem.

**Problem 2.1** (R:2.2\*). A complex number  $\gamma$  is **algebraic** iff  $\gamma$  is a root to a polynomial with integer coefficients. Prove that there are complex numbers that are not algebraic.

The following are two fun facts just beyond what we got to in MAT-513:

Let  $\mathbb{A} \subset \mathbb{C}$  be the set of algebraic numbers.

1.  $\mathbb{A}$  is a field.
2.  $\mathbb{A}$  is algebraically closed, that is, if  $\alpha$  is a root of a polynomial in  $\mathbb{A}[x]$ , then  $\alpha \in \mathbb{A}$ . So in the definition of algebraic numbers you can use any ring of coefficients  $R$ ,  $\mathbb{Z} \subseteq R \subseteq \mathbb{A}$ .

**Problem 2.2** (B:2.2:4). This problem is concerning relative topology.

- (a) Suppose  $E$  is an open subset of a metric space  $X$ , and  $F$  is a subset of  $E$ . Show that  $F$  is open relative to  $E$  iff  $F$  is open.
- (b) Suppose  $E$  is a closed subset of a metric space  $X$ , and  $F$  is a subset of  $E$ . Show that  $F$  is closed relative to  $E$  iff  $F$  is open.
- (c) Suppose  $E$  is an open subset of a metric space  $X$ , and  $F$  is a subset of  $E$  that is closed relative to  $E$ . Show that  $F$  need not be open or closed in  $X$ .
- (d) Suppose  $E$  is a closed subset of a metric space  $X$ , and  $F$  is a subset of  $E$  that is open relative to  $E$ . Show that  $F$  need not be open or closed in  $X$ .

**Problem 2.3** (B:2.2:20). Here you look at the Kuratowski game discussed in the podcast.

- (a) Show that  $\text{Cl}(\text{Cl}(E)) = \text{Cl}(E)$  and  $\text{Int}(\text{Int}(E)) = \text{Int}(E)$ .
- (b)  $\text{Int Cl}(E) = \text{Int}(\text{Cl}(\text{Int}(\text{Cl}(E))))$ .
- (c) Show that you can get at most seven different sets from applying the operations of closure and interior repeatedly to a fixed set  $E$ .
- (d) Show that there is  $E \subset \mathbb{R}$  so that all seven possible sets are realized.

**Bonus:** If you also throw in the operation of complement, then there are at most 14 possible sets you can get starting from a fixed set  $E$ .

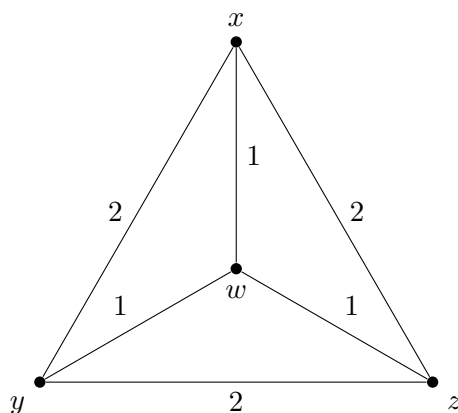
**Problem 2.4** (B:2.2:14). This investigates the  $p$ -adic metric on  $\mathbb{Z}$ . Let  $n > 1$  be an integer. Define  $d_p(s, t) = n^{-e_n(s-t)}$  where  $e_n(m) = k$  iff  $n^k \mid m \wedge n^{k+1} \nmid m$ , so  $e_n(m)$  reports the maximal number of times  $m$  can be evenly divided by  $n$ .

Show that  $d_n : \mathbb{Z} \times \mathbb{Z} \rightarrow [0, \infty)$  is a metric and in fact an **ultra-metric**, i.e.,  $d_n(p, q) \leq \max\{d_n(p, r), d_n(r, q)\}$ .

**For Fun:** Show that all  $p$ -adic triangles are isosceles. This is a geometric idea; what corresponding topological fact is leading to this?

$p$ -adics are very important and interesting. [Here is a great video introduction to p-adics](#), which discusses applications to solving Diophantine equations.

**Problem 2.5** (B:2.2:18). Let  $X$  be a 4-element set. Consider a graph on the four nodes  $\{x, y, z, w\}$  with edges and edge weights as shown.



- Verify that  $d_X : X \times X \rightarrow \mathbb{R}$  is a metric where  $d(x, y) = d(x, z) = d(y, z) = 2$  and  $d(w, x) = d(w, y) = d(w, z) = 1$ .
- Show that there is no  $F : X \rightarrow \mathbb{R}^k$  that is distance preserving, that is,  $d_X(u, v) = d_{\mathbb{R}^k}(f(u), f(v))$ .
- The example here is a 4-point space that cannot be *isomorphically* embedded in  $\mathbb{R}^k$  for any  $k$ , yet every 3-point subspace can be isomorphically embedded in  $\mathbb{R}^k$ . Can you find a 5-point space that cannot be isomorphically embedded in  $\mathbb{R}^k$  and yet all of whose 4-point subspaces can be isomorphically embedded in  $\mathbb{R}^k$  for sufficiently large  $k$ .

**Problem 2.6** (B:2.2:19). Let  $X$  be a countable (countably infinite) metric space. show that there are uncountably many open subsets of  $X$ .

- Let  $D = \{p_i \mid i \in \mathbb{N}\}$  be a countable dense set. Find reals  $r_i$  so that the neighborhoods  $N_{r_i}(p_i)$  of  $p_i$  are mutually disjoint. That is  $N_{r_i}(p_i) \cap N_{r_j}(p_j) = \emptyset$ .
- Associate with every  $\alpha \in 2^{\mathbb{N}}$  the open set  $O_\alpha = \bigcup_{\{i \mid \alpha(i)=1\}} N_{r_i}(p_i)$ . Show that this map is an injection of  $2^{\mathbb{N}}$  into the open sets of  $X$ .
- (b) shows that there are at least continuum many open subsets of  $X$ , could the cardinality be even larger?

**Problem 2.7** (R:2.26). Let  $X$  be a metric space in which every infinite set,  $A$ , has a limit point in  $A$ . Show that  $X$  is compact. (Follow the hints in Rudin.)

**This is a very important property.** We will see that for metric spaces, this gives the equivalence of " $X$  compact" with " $X$  is sequentially compact."

**Problem 2.8** (B:2.3:3 - Compact sets get crowded!). Let  $X$  be a compact metric space and let  $\epsilon > 0$ . Show that there is an  $N$  so that for any  $m \geq N$ , if  $p_1, \dots, p_m$  are  $m$  distinct points in  $X$ , then at least two of them are within  $\epsilon$  of each other, that is, there are  $i, j$  with  $d(p_i, p_j) < \epsilon$ .

**Problem 2.9** (B:2.5:1 - The usual definition of connected.). (a) Show that  $E \subsetneq X$  is connected (as per Rudin) iff there is  $\emptyset \neq A \subsetneq E$  that is **clopen**, i.e., closed and open.

(b) Show that  $E \subsetneq X$  is connected (as per Rudin) iff there are two non-empty **relatively open subsets** of  $E$  that **cover**  $E$ .

**Note:** A space  $X$  is connected iff the clopen sets consist of exactly  $X$  and  $\emptyset$ .

**Problem 2.10** (This is a mix of problems.). Consider the following collections of open subsets of  $\mathbb{Q}$ . (1) Verify that each gives a topology on  $\mathbb{Q}$  and (2) compute for each,  $\text{Cl}((0, \sqrt{2}))$ , (3) show that the space is compact, or not, (4) show that the space is connected or not, and finally (5) how are these related to the standard relative topology that  $\mathbb{Q}$  inherits from  $\mathbb{R}$ ?

- (a) *The left-ray topology.* Take  $\mathcal{O}$  to be the set of all right-rays of the form  $(q, \infty)$  for  $q \in \mathbb{Q} \cup \{\infty\}$ .
- (b) *The co-finite topology.* Take  $\mathcal{O}$  to be subsets of  $\mathbb{Q}$  whose complements are finite.
- (c) *The order topology.* A set  $A$  is open in this space iff for all  $q \in A$ , there is  $q_1 < q < q_2$  in  $\mathbb{Q}$  so that  $(q_1, q_2) \cap \mathbb{Q} \subseteq A$ .